

Unit:8

Security Management

Prepared by

Er. Binod Kumar Rajbhar

BE-Computer

ME-Computer

Computer security

What is computer security?

- ✓ Computer security basically is the protection of computer systems and information from harm, theft, and unauthorized use.
- ✓ It is the process of preventing and detecting unauthorized use of your computer system.
- ✓ There are various types of computer security which is widely used to protect the valuable information of an organization.

Types of Computer Security

- ✓ **Information security** is securing information from unauthorized access, modification & deletion
- ✓ **Application Security** is securing an application by building security features to prevent from Cyber Threats such as SQL injection, DoS attacks, data breaches and etc.
- ✓ **Computer Security** means securing a standalone machine by keeping it updated and patched
- ✓ **Network Security** is by securing both the software and hardware technologies
- ✓ **Cybersecurity** is defined as protecting computer systems, which communicate over the computer networks

Security Problems

- ❖ Security must consider external environment of the system and protect the system resources.
- ❖ There are various security problems in operating system out of them major security problems are listed as below:
 - ✓ User authentication
 - ✓ Program threats
 - ✓ System threats

User Authentication

- ✓ Authentication refers to identifying each user of the system and associating the executing programs with those users.
- ✓ It is the responsibility of the operating system to create a protection system which ensures that a user who is running a particular program is authentic.

1. Password:

- User need to enter a registered username and password with operating system to login into the system.

2.Password Vulnerabilities

- ✓ Vulnerability is a cyber security term that refers to a flaw in a system that can leave it open to attack.
- ✓ Vulnerability may also refer to any type of weakness in a computer system itself, in a set of procedure or in anything that leaves information security exposed to a threats.
- ✓ Strong protection starts with strong passwords, use a variety of lowercase and uppercase letters, numbers, characters and symbols etc. and be sure to change them every few months.
- ✓ Never use combinations that include personal information or are easy to guess, like the website name, your name, birthday or social security number.
- ✓ Don't use the same password for more than one account , it seems like the easy choice but it come with risks.
- ✓ If a hacker does get into one of your accounts, then they will be able to access all the other ones with the same login.

2.Encrypted Password

- ✓ Encryption is the process of encoding a message or information in such a way that only authorized parties can access it and those who are not authorized can not.
- ✓ Using one –way encryption formats user passwords may be encrypted and stored in the directory, which prevents clear password form being accessed by any users including the system administrators.
- ✓ Using two-way encryption formats , password are encrypted while stored in the database and decrypted when returned to an authorized client.

One time password and Biometrics Password

One time password:

- ✓ one time password provide additional security along with normal authentication.
- ✓ One time password system, a unique password is required every time user tries to login into the system.
- ✓ Once a one time password is used, then it cannot be used again. Use

One time password

- ✓ Along with standard authentication, one-time passwords give an extra layer of security.
- ✓ Every time a user attempts to log into the One-Time Password system, a unique password is needed.
- ✓ Once a one-time password has been used, it cannot be reused.
- ✓ One-time passwords may be implemented in several ways.

1. Secret Key

- The user is given a hardware device that can generate a secret id that is linked to the user's id. The system prompts for such a secret id, which must be generated each time you log in.

2. Random numbers

- Users are given cards that have alphabets and numbers printed on them. The system requests numbers that correspond to a few alphabets chosen at random.

3. Network password

- Some commercial applications issue one-time passwords to registered mobile/email addresses, which must be input before logging in.

Biometrics Password

- ✓ Biometrics authentication is a security process that relies on the unique biological characteristics of an individual
- ✓ Common biometric authentication methods include:
 - **Facial recognition**—matches the different face characteristics of an individual trying to gain access to an approved face stored in a database. Face recognition can be inconsistent when comparing faces at different angles or comparing people who look similar, like close relatives.
 - **Fingerprint scanners**—match the unique patterns on an individual's fingerprints. Some new versions of fingerprint scanners can even assess the vascular patterns in people's fingers. Fingerprint scanners are currently the most popular biometric technology for everyday consumers, despite their frequent inaccuracies. This popularity can be attributed to iPhones.
 - **Speaker Recognition** —also known as voice biometrics, examines a speaker's speech patterns for the formation of specific shapes and sound qualities. A voice-protected device usually relies on standardized words to identify users, just like a password.
 - **Eye scanners**—include technologies like iris recognition and retina scanners. Iris scanners project a bright light towards the eye and search for unique patterns in the colored ring around the pupil of the eye. The patterns are then compared to approved information stored in a database. Eye-based authentication may

User Authorizations

- ✓ Authorization is the function of specifying access rights/ privileges to resources , which is related to information security and computer security in general and to access control in particular.
- ✓ Authorization is a process by which a server determines if the client has permission to use a resource or access a file.
- ✓ Authorization is usually coupled with authentication so that the server has some concept of who the client is that is requesting access.
- ✓ The type of authentication required for authorization may vary; passwords may be required in some cases but not in others.
- ✓ In some cases, there is no authorization; any user may be use a resource or access a file simply by asking for it. Most of the web pages on the Internet require no authentication or authorization.

Program Threats

- ✓ The operating system's processes and kernel carry out the specified task as directed.
- ✓ Program Threats occur when a user program causes these processes to do malicious operations.
- ✓ The common example of a program threat is that when a program is installed on a computer, it could store and transfer user credentials to a hacker.
- ✓ There are various program threats, one of them are as follows

Program Threats

✓ Trojan Horse:

This type of application captures user login credentials. It stores them to transfer them to a malicious user who can then log in to the computer and access system resources.

✓ Logic Bomb

A logic bomb is a situation in which software only misbehaves when particular criteria are met; otherwise, it functions normally.

✓ Trap Door

A trap door is when a program that is supposed to work as expected has a security weakness in its code that allows it to do illegal actions without the user's knowledge.

Stack and Buffer Overflow

System Threats

- ✓ System threats are described as the misuse of system services and network connections to cause user problems.
- ✓ These threats may be used to trigger the program threats over an entire network, known as program attacks.
- ✓ System threats make an environment in which OS resources and user files may be misused.
- ✓ There are various system threats ,some of them are as follows:

1. Port Scanning

- ✓ It is a method by which the cracker determines the system's vulnerabilities for an attack.
- ✓ It is a fully automated process that includes connecting to a specific port via TCP/IP.
- ✓ To protect the attacker's identity, port scanning attacks are launched through Zombie Systems, which previously independent systems now serve their owners while being utilized for such terrible purposes.

2.Computer Virus:

- Computer virus is a program, script or macro designed to cause damage, steal personal information, modify data, and send email, display message or some combination of these action.
- A computer virus is a malicious program that self-replicates by copying itself to another program.
- In other words, the computer virus spreads by itself into other executable code or documents.
- **A computer virus attack can produce a variety of symptoms, here are some of them:**
 - ✓ Changes to our homepage
 - ✓ Mass email being sent from our email account
 - ✓ Frequent crashes
 - ✓ Unusually slow computer performance
 - ✓ Unknown programs that start up when we turn on our computer
 - ✓ Unusual activities like password changes

Virus can be Spread:

- ✓ File sharing
- ✓ Downloading files or software
- ✓ Instant messaging
- ✓ Floppy disk, pen drive
- ✓ Web page.

Tips to Protect Your Computer From Viruses and Malware

- ✓ Keep your software up to date.
- ✓ Don't click on links within emails.
- ✓ Use free antivirus software.
- ✓ Back up your computer.
- ✓ Use a strong password.
- ✓ Use a firewall.
- ✓ Minimize downloads.
- ✓ Use a pop-up blocker.

3. Worm

- ✓ A computer worm is a malicious, self-replicating software program(popularly termed as malware) which affects the function of software and hardware programs.
- ✓ The worm is a process that can choke a system's performance by exhausting all system resources.
- ✓ A Worm process makes several clones, each consuming system resources and preventing all other processes from getting essential resources.
- ✓ Worm processes can even bring a network to a halt.

4. Denial-of-service

- ✓ A denial-of-service (DoS) attack is a type of cyber attack in which a malicious actor aims to render a computer or other device unavailable to its intended users by interrupting the device's normal functioning.
- ✓ A **Denial-of-Service (DoS) attack** is an attack meant to shut down a machine or network, making it inaccessible to its intended users.
- ✓ DoS attacks accomplish this by flooding the target with traffic, or sending it information that triggers a crash.
- ✓ In both instances, the DoS attack deprives legitimate users (i.e. employees, members, or account holders) of the service or resource they expected.

- ✓ There are two general methods of DoS attacks: flooding services or crashing services.
- ✓ Flood attacks occur when the system receives too much traffic for the server to buffer, causing them to slow down and eventually stop.
Popular flood attacks include:

Buffer overflow attacks –

- ✓ The most common DoS attack. The concept is to send more traffic to a network address than the programmers have built the system to handle.
- ✓ It includes the attacks listed below, in addition to others that are designed to exploit bugs specific to certain applications or networks